**8.** *Staphylococcus aureus* is a bacterium which is a common cause of infection in human wounds.

*S. aureus* has become resistant to antibiotics.

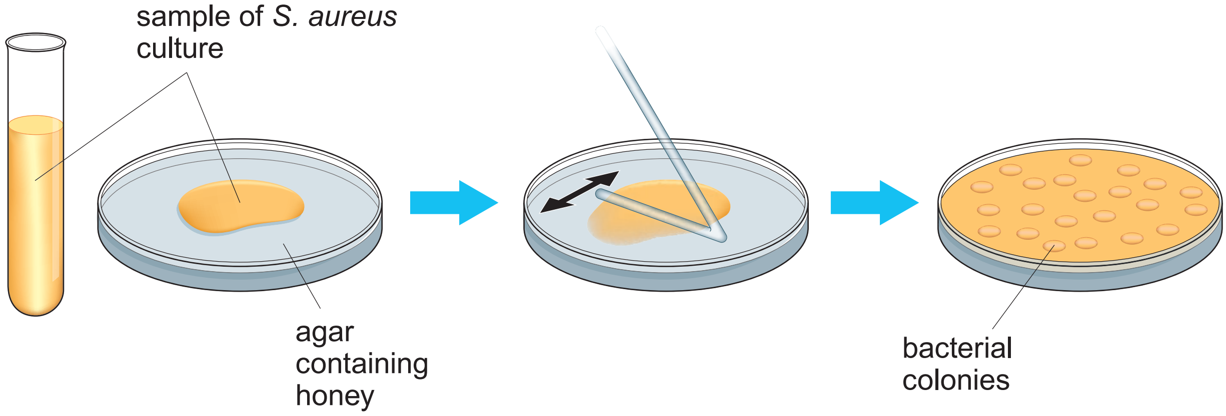
Honey has been used as a wound dressing for thousands of years. Scientists carried out a laboratory investigation to study the effect of honey on the growth of *S. aureus*.

A series of agar plates containing different honey concentrations was prepared.
A culture containing *S. aureus* was transferred to each of the agar plates.

**1** 0.3 $mm^3$ of *S. aureus* culture poured onto honey agar

**2** Spread sample evenly over the surface

**3** Agar containing honey at 3% concentration at 24 hours.




The plates were incubated for 24 hours and then inspected for the presence of bacterial colonies. The results are shown below.

| Concentration of Honey (%) | Bacterial colonies present (✓) or absent (×) |
|---|---|
| 0 | ✓ |
| 1 | ✓ |
| 2 | ✓ |
| 3 | ✓ |
| 4 | × |
| 5 | × |
| 6 | × |
| 7 | × |
| 8 | × |
| 9 | × |
| 10 | × |



Examiner only

*(a)* State the link between the number of bacterial colonies present on the agar in diagram **3** at 24 hours and the number of bacteria in the original sample. [1]

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*(b)* (i) Suggest the temperature at which the agar plates would have to be incubated in this investigation. Give a reason for your answer. [2]

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(ii) State **two** basic aseptic techniques that should have been followed in preparing the plates shown in the diagram. [2]

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*(c)* Using the diagrams, calculate the number of bacteria present in **1 cm$^3$** of the original *S. aureus* culture. **Present your answer in standard form**.
(Note 1 cm$^3$ = 1000 mm$^3$) [3]

Number of bacteria present = .......... bacteria / cm$^3$

Examiner only

*(d)* (i) Using the table on page 20, state the minimum concentration of honey required to stop the growth of bacteria for 24 hours. [1]

Minimum concentration = .................................................... %

(ii) Suggest how scientists could develop their investigation to get a more accurate value for the minimum concentration of honey required to stop the growth of bacteria for 24 hours. [1]

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*(e)* State the **three** pre-clinical stages that would have to be carried out in order to develop honey as a new medicine for the treatment of wounds. [3]

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